Listing 9.4 `aggregate.py`

```
@tool
def aggregate(processed_results: List[str]):
    items = [item for sublist in processed_results
              ↪ for item in sublist]     #1

    aggregated = {}

    for item in items:
        for key, value in item.items():
            if key == 'title':
                continue

            if isinstance(value, (float, int)):      #2
                if key in aggregated:
                    aggregated[key] += value
                else:
                    aggregated[key] = value

    for key, value in aggregated.items():      #3
        value = value / len(items)
        log_metric(key=key, value=value)     #4
        aggregated[key] = value

    return aggregated
```

#1 The input is a list of lists; flatten to a list of items.
#2 Checks to see if the value is numeric and accumulates scores for each criterion key
#3 Loops over aggregated criterion scores
#4 Logs the criterion as a metric

The result of the aggregations will be a summary score for each criterion and the average score. Since the evaluation/grounding flow is separate, it can be run over any recommendation run we perform. This will allow us to use the batch run results for any variation to compare results.

We can run the grounding flow by opening `flow.dag.yaml` in the visual editor and clicking Batch (beaker icon). Then, when prompted, we select an existing run and then select the run we want to evaluate, as shown in figure 9.25. This will open a YAML file with the Run link at the bottom, as we've seen before. Click the Run link to run the evaluation.